

Kenneth Shaw

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EDUCATION

- **Carnegie Mellon University Robotics Institute** Aug 2023 - Present
Ph.D in Robotics
Advised by Prof. Deepak Pathak
Pittsburgh, PA
- **Carnegie Mellon University Robotics Institute** Aug 2020 - Aug 2023
M.S. in Robotics
Advised by Prof. Deepak Pathak
Thesis: Designing and Teaching Dexterous Robot Hands
Pittsburgh, PA
- **Georgia Institute of Technology** Aug 2017 - May 2020
B.S. in Computer Engineering (Minor: CS in Intelligence)
GPA 3.94/4.00
Advised by Prof. Sonia Chernova and Prof. Harish Ravichandar
Atlanta, Georgia

PUBLICATIONS

[Google Scholar](#)

- [1] *IFG: Internet-Scale Functional Grasping*
R. Liu, M. Li, **K. Shaw**, D. Pathak
In Submission
- [2] *LEAP Hand V2 Advanced: Dexterous, Low-cost Hybrid Rigid-Soft Hand for Robot Learning*
K. Shaw, D. Pathak
IEEE Humanoids 2025
Best Paper Presentation Award at RSS 2024 Dex Manip Workshop
- [3] *Deep Reactive Policy: Learning Reactive Manipulator Motion Planning for Dynamic Environments*
J. Yang, J.J. Liu, Y. Li, Y. Khaky, **K. Shaw**, D. Pathak
Conference on Robot Learning 2025
Best Paper Award at LeaPRiDE IROS Workshop
- [4] *FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning*
J.J. Liu, Y. Li, **K. Shaw**, T. Tao, R. Salakhutdinov, D. Pathak
Robotics: Science and Systems 2025
- [5] *DexWild: Dexterous Human Interactions for In-the-Wild Robot Policies*
T. Tao, M. Srirama, J.J. Liu, **K. Shaw**, D. Pathak
Robotics: Science and Systems 2025
Best Paper Award at CoRL EgoAct Workshop
- [6] *Demonstrating LEAP Hand V2: Low-Cost, Easy-to-Assemble, High-Performance Hand for Robot Learning*
K. Shaw, D. Pathak
Robotics: Science and Systems 2025
- [7] *Bimanual Dexterity for Complex Tasks*
K. Shaw*, Y. Li*, J. Yang, M. K. Srirama, R. Liu, H. Xiong, R. Mendonca, D. Pathak
Conference on Robot Learning 2024
- [8] *Adaptive Mobile Manipulation for Articulated Objects in the Open World*
H. Xiong, R. Mendonca, **K. Shaw**, D. Pathak
Arxiv 2024
- [9] *SPIN: Simultaneous Perception, Interaction and Navigation*
S. Uppal, A. Agarwal, H. Xiong, **K. Shaw**, D. Pathak
Computer Vision and Pattern Recognition 2024
- [10] *DEFT: Dexterous Fine-Tuning for Real-World Hand Policies*
A. Kannan*, **K. Shaw***, S. Bahl, P. Mannam, D. Pathak
Conference on Robot Learning 2023

- [11] *DASH: A Framework for Designing Anthropomorphic Soft Hands through Interaction*
P. Mannam*, **K. Shaw***, D. Bauer, J. Oh, D. Pathak, N. Pollard
IEEE Humanoids 2023
Best Oral Paper Award Finalist (Top 3)
- [12] *Dexterous Functional Grasping*
A. Agarwal, S. Uppal, **K. Shaw**, D. Pathak
Conference on Robot Learning 2023
- [13] *Learning Dexterity from Human Hand Motion in Internet Videos*
K. Shaw, S. Bahl, A. Sivakumar, A. Kannan, D. Pathak
IJRR Special Issue 2023
Featured on Front Cover, April 2024
- [14] *LEAP Hand: Low-Cost, Efficient, and Anthropomorphic Hand for Robot Learning*
K. Shaw, A. Agarwal, D. Pathak
Robotics: Science and Systems 2023
- [15] *VideoDex: Learning Dexterity from Internet Videos*
K. Shaw*, S. Bahl*, D. Pathak
Conference on Robot Learning 2022
- [16] *Robotic Telekinesis: Learning a Robotic Hand Imitator by Watching Humans on YouTube*
A. Sivakumar*, **K. Shaw***, D. Pathak
Robotics: Science and Systems 2022
Best Paper Award Finalist in Scaling Robot Learning Workshop
- [17] *Predicting Individual Human Performance in Human-Robot Teaming*
J. Kolb, M. Kishore, **K. Shaw**, H. Ravichandar, S. Chernova
RO-MAN 2021
- [18] *Strata: Unified Framework for Task Assignments in Large Teams of Heterogeneous Agents*
H. Ravichandar, **K. Shaw**, S. Chernova
AAMAS — JAAMAS Track, 2021
- [19] *Approximated Dynamic Trait Models for Heterogeneous Multi-Robot Teams*
G. Neville, H. Ravichandar, **K. Shaw**, S. Chernova
International Conference on Intelligent Robots and Systems 2020
- [20] *Quantum Computing: Evaluating Potential Quantification of Projective Psychological Test Scoring*
D. Davis, **K. Shaw**, S. Rizvi, M. Davis
MODSIM WORLD 2019
- [21] *Feasibility and Usability of MentorPal, a Framework for Rapid Development of Virtual Mentors*
B. Nye, D. Davis, S. Rizvi, K. Carr, W. Swartout, R. Thacker, D. Cobbins, **K. Shaw**
Journal of Research on Technology in Education (JTRE), 2019

EXPERIENCE

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| <ul style="list-style-type: none"> • Nvidia Robotics
<i>PhD Research Intern - Robotics Sim2Real</i> • Robotics Institute Summer Scholar: Carnegie Mellon Univ.
<i>NSF REU Student Researcher</i> • University of Southern California: Institute for Creative Technologies
<i>NSF REU Student Researcher</i> | <p><i>June 2024 - April 2025</i>
Seattle, Washington and Remote</p> <p><i>June 2019 - Sept. 2019</i>
Pittsburgh, Pennsylvania</p> <p><i>May 2018 - Sept. 2018</i>
Los Angeles, California</p> |
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HONORS AND AWARDS

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| <ul style="list-style-type: none"> • NSF Graduate Research Fellowship Recipient • Best Paper Award at LeaPRiDE IROS Workshop • Best Paper Award at CoRL EgoAct Workshop • Best Paper Presentation Award at RSS Dexterous Manipulation Workshop | <p>2020–2024</p> <p>2025</p> <p>2025</p> <p>2024</p> |
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• Best Oral Paper Finalist, IEEE Humanoids (Top 3)	2023
• Featured IJRR Article (Front Cover) in Volume 43, Issue 4	2024
• Best Paper Award Finalist in Scaling Robot Learning Workshop	2022
• Warren Batts & Austin Brown Innovation Award Scholarship	2019
• Georgia Tech Faculty Honors	2017–2020
• Georgia Tech Dean’s List (All Semesters)	2017–2020
• PennApps Hackathon Top 30 Winner: TensorFlow image recognition to facilitate recycling	2018
• HackMIT Sia API Challenge 1st Place: Used the Sia Blockchain for ad-supported file storage	2018

SELECTED INVITED TALKS

• R-PAD Lab at CMU Full Talk — <i>LEAP Hand</i>	November 2025
• STAR Lab at Georgia Institute of Technology — <i>LEAP Hand v2</i>	November 2025
• Open-Source Hardware in the Era of Robot Learning Workshop at CoRL — <i>LEAP Hand v2</i>	September 2025
• Apple Robotics (FAR) — <i>Robot Hand Teleoperation</i>	June 2025
• Amazon Robotics — <i>LEAP Hand</i>	April 2025
• Google Robotics — <i>Bimanual Dexterity for Complex Tasks</i>	August 2024
• Sony Robotics (RAP) — <i>Learning Dexterity from Human Data</i>	April 2024
• Humanoid Generalization (HuGe) Lab at KAIST — <i>LEAP Hand</i>	April 2024
• Robotics Institute Summer Scholars — <i>Deep Learning Crash Course</i>	June 2022

SELECTED INVITED DEMOS

• RSS Demo 2025 - <i>Demonstrating LEAP Hand v2: Low-Cost, Easy-to-Assemble, High-Performance Hand for Robot Learning</i>	July 2025
• RSS Demo 2024 - <i>Demonstrating LEAP Hand v2: Dexterous, Low-cost Anthropomorphic Hybrid Rigid Soft Hand for Robot Learning</i>	July 2024
• RSS Demo 2023 - <i>LEAP Hand (Systems Paper)</i>	July 2023
• CoRL Demo - <i>LEAP Hand</i>	November 2023/4/5
• ICRA Demo - <i>LEAP Hand</i>	June 2025

MENTORSHIP

Students I have mentored in Deepak’s lab:

- Jason Liu (*Current Ph.D*)
- Andrew Wang (*Current Ph.D*)
- Sri Anumakonda (*Current B.S.*)
- Muxin Liu (*Current B.S.*)
- Tony Tao (*M.S. 2025*)
- Mohan Kumar Srirama (*M.S. 2025*)
- Jiahui Yang (*M.S. 2025*)
- Mingxuan Li (*B.S. 2025*)
- Kehlani Fay (*B.S. 2024*)
- Haoyu Xiong (*M.S. 2024*)
- Shagun Uppal (*M.S. 2024*)
- Yulong Li (*M.S. 2024*)

ACADEMIC SERVICE

- Dexterous Manipulation: Learning and Control with Diverse Modalities [CoRL]
Lead Organizer September 2025
- Dexterous Manipulation: Learning and Control with Diverse Data [RSS] June 2025
- Handy Moves: Dexterity in Multi-Fingered Hands [ICRA] May 2025
- Conference Reviewer: ICRA (2023, 2024, 2025), IROS (2024, 2025), CoRL (2024)
- Journal Reviewer: RA-L (2023, 2024, 2025), T-RO (2024, 2025)
- AI4ALL at CMU: Mentorship of high school students in AI projects Summer 2021
- First Robotics Competition (FRC) Mentor 2018 - Current

TEACHING EXPERIENCE

- **Teaching Assistant:** Spring 2025
Introduction to Robot Learning (16-831) with Prof. Guanya Shi Carnegie Mellon University
- **Teaching Assistant:** Spring 2024
Visual Learning and Recognition (16-824) with Prof. Deepak Pathak Carnegie Mellon University

SELECT MEDIA COVERAGE

- DexWild: Dexterous Human Interactions for In-the-Wild Robot Policies
RoboPapers (Chris Paxton) Fall 2025
- LEAP Hand: Low-Cost, Efficient, and Anthropomorphic Hand for Robot Learning
The Daily Beast (How You Can Make This Powerful Robotic Hand at Home) Fall 2023
- Adaptive Mobile Manipulation In the Open World
New Scientist, Nature Briefings, IOT World Today, TechXplore Spring 2022
- Robotic Telekinesis: Learning a Robotic Hand Imitator by Watching Humans on Youtube
TechXplore, The Voice of America, Weights& Biases, TwoMinutePapers Spring 2022